



Illuminating
ENGINEERING SOCIETY

**APPROVED METHOD:
GUIDE TO GONIOMETER
MEASUREMENTS AND TYPES, AND
PHOTOMETRIC COORDINATE SYSTEMS**
AN AMERICAN NATIONAL STANDARD



ANSI/IES LM-75-19

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AND PHOTOMETRIC COORDINATE SYSTEMS
AN AMERICAN NATIONAL STANDARD**

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Foreword

This document is an update of LM-75-2001, which was reaffirmed in 2012. The document has been updated to reflect current use of goniophotometers in industry, including calibration methods and methodologies for correction of stray light. Updated in this document are definitions of goniophotometer types and coordinate systems for photometric measurement of light sources. This document supersedes any angular conventions presented in LM-72-1997. This document does not cover near-field goniometers.

1.0 Introduction and Scope

1.1 Introduction

A goniophotometer is a photometer for measuring the directional light distribution characteristics of sources, luminaires, media and surfaces. A goniophotometer records photometric readings at a series of spherical coordinates to define a web of photometric data surrounding the item under test. Goniophotometric data are presented using various angular coordinate systems, and acquired using instruments of various constructions. In addition to goniophotometers, goniospectroradiometers exist, which use three- or four-channel colorimeters to measure color properties at a series of spherical coordinates, and goniospectroradiometers exist, which use array spectrometers to measure spectral properties at a series of spherical coordinates. Historically, each type of goniophotometer was developed to measure a light source used in a specific application. Conventionally, measurement results for each type of light source were presented using a coordinate system that matched the specific application. Many current IES documents mention goniophotometers and coordinate systems. This document offers a complete explanation and provides harmonious definitions.

For luminaires, goniophotometers are used to measure luminous intensity distributions using either the relative or absolute photometry method. Using relative photometry, the luminous intensity in each direction is derived by normalizing the indicated

luminous intensity of the luminaire and then scaling it to the rated initial lumens of the bare lamp(s) used in the luminaire; thus, absolute photometric calibration of the goniophotometer is not necessary using this method. However, relative photometry does not work for certain solid-state lighting (SSL) products, where the source(s) cannot be separated from the luminaire. For such products, absolute photometry, which requires a calibrated goniophotometer, shall be used.

This document provides definitions of goniophotometer types, spherical coordinate systems, and a general guide to goniophotometer calibration. Definitions presented herein are generally consistent with those in corresponding CIE publications [CIE 102¹ and CIE 121²]. Differences or inconsistencies are noted.

1.2 Scope

This document provides definitions of spherical coordinate systems and goniophotometer types used to measure light sources. It does not address the use of goniophotometers to measure media or surfaces. The operating principles behind each type of goniophotometer are addressed, and a general guide to goniophotometer calibration, stray light elimination, and stray light correction is presented. A methodology for zeroing data when measuring upward light is also provided. In addition, the correspondences between relevant IES and CIE definitions are described.

2.0 Normative References

Illuminating Engineering Society. ANSI/IES RP-16-17, Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2017. Online: www.ies.org/standards/definitions/. (Accessed 2019 Jul 26).

3.0 Definitions, Including Spherical Coordinates

3.1 apolar angle

In a spherical coordinate system,^{3,4} an angle measured from the positive pole; with the circumpolar angle, one of two

angular coordinates used to specify a unique direction. For any given direction, the apolar angle is the complement of the elevation angle. (See **Figure 5-2, Section 5.1**.)

Note: The apolar angle is commonly identified as the zenith angle, but the term *zenith* and its antonym, *nadir*, have generally been avoided in this document, due to possible misleading associations with the direction of gravity.

3.2 attitude, application

The orientation in which a light source is operated in the field.

Note: If not otherwise indicated, application attitude is assumed to match design attitude.

3.3 attitude, design

The orientation in which a light source is designed to operate, as specified by the manufacturer.

3.4 attitude, measurement

The orientation in which a light source is measured.

Note: If not otherwise indicated, measurement attitude is assumed to match design attitude.

3.5 azimuth angle

(See **Section 3.9 circumpolar angle**.)

3.6 CIE (A, α) Coordinate System

A left-handed spherical coordinate system in a source frame of reference, in which the third axis of the light source is aligned with the pole of the coordinate system, while the direction of the receiver is given by an elevation angle (α) and a circumpolar angle (A) measured from the first axis of the light source.

Note: This coordinate system is equivalent to the IES Coordinate System A.

3.7 CIE (B, β) Coordinate System

A right-handed spherical coordinate system in a source frame of reference, in which the second axis of the light source is aligned with the pole of the coordinate system, while the direction of the receiver is given by an elevation angle (β) and a circumpolar angle (B) measured from the first axis of the light source.

Note: This coordinate system is equivalent to the IES Coordinate System B.

3.8 CIE (C, γ) Coordinate System

A left-handed spherical coordinate system in a source frame of reference, in which the pole is a specified test axis for the light source, while the direction of the receiver is given by an apolar angle (γ) and a circumpolar angle (C) measured from the half-plane containing the second axis of the light source. (See also **Section 3.33 test axis**.)

Note 1: In the typical case of zero tilt, the specified test axis is identical with the first axis of the light source, and the second axis of the light source is simply aligned with the instrument's datum axis ($C=0^\circ$, $\gamma=90^\circ$). (See also **Section 3.35 tilt, measurement**.)

Note 2: This coordinate system is equivalent to the IES Coordinate System C.

3.9 circumpolar angle (azimuth angle)

In a spherical coordinate system, an angle measured about the polar axis from a specified datum half-plane; with the planar angle, one of two angular coordinates used to specify a unique direction. In **Figure 5-2 (Section 5.1)**, this coordinate (ϕ), is the angle between the datum half-plane and the measurement plane. The sign of ϕ depends on whether the system is right- or left-handed.

Note: The circumpolar angle may also be described as the aplanar angle, the angle from the (datum) plane, or the angle to the (measurement) plane.

3.10 coordinate system

A system of numerical indices by which the position or direction of a point within a given frame of reference may be expressed.

Note: A three-dimensional coordinate system may be Cartesian (rectilinear), spherical, or other. An abridged spherical coordinate system is used in goniophotometry.

3.11 datum axis

The axis that identifies the starting point for measuring the circumpolar angle in the equatorial plane (see **Figure 5-2**). The location of the datum axis is dependent on the application and should always be clearly defined.

3.12 datum half-plane

The half-plane that includes the positive half of the datum axis, from which the circumpolar angle is measured (see **Figure 5-2, Section 5.1**).

3.13 elevation angle

In a spherical coordinate system, an angle measured from the equatorial plane (see **Figure 5-2, Section 5.1**); with the circumpolar angle, one of two angular coordinates used to specify a unique direction. For any given direction, the elevation angle is the complement of the apolar angle.

3.14 first axis (of a light source)

An axis through the photometric center of a light source, corresponding to the design attitude, or standard orientation, of the source.

Note: The first axis of the light source generally indicates the direction in which the light source would typically be aimed. For non-directional sources, the definition of this axis may be arbitrary.

3.15 frame of reference

A Euclidean space, represented by a set of orthogonal axes, motionless with respect to a given object or observer, which may serve as the basis for the definition of a coordinate system.

3.16 frame of reference, application

A frame of reference defined with respect to the field environment in which a light source is installed and operated.

Note: If, upon installation, the source axes are aligned with those of the application environment, the application frame of reference is equivalent to the source frame of reference.

3.17 frame of reference, goniometer

A frame of reference defined with respect to the optical axis of detection and the axes of rotation of a given goniophotometer system.

Note: In the goniometer frame of reference, the source rotates during measurement, while the axis of detection remains fixed.

3.18 frame of reference, laboratory

A frame of reference defined with respect to the laboratory environment in which a light source is tested.

Note: Depending on the goniophotometer design, both the light source and the detector may move within the laboratory frame during measurement.

3.19 frame of reference, source

A frame of reference defined with respect to a given light source.

Note: In the source frame of reference, the detector moves with respect to the source during measurement, while the source remains fixed.

3.20 half-plane of data

The set of data points acquired by rotating a goniophotometric detector from pole to pole within a half-plane (180° of rotation) at a specified angular interval.

3.21 half-plane, goniometric

Half of a plane that contains, and is bisected by, the polar axis. A half-plane contains the full range of directions at a given circumpolar angle, corresponding to apolar angles ranging from 0° to 180°, or elevation angles ranging from -90° to 90°. (See **Figure 5-2, Section 5.1**.)

3.22 handedness (of a coordinate system)

The direction of positive circumpolar rotation, given the orientation of the polar and datum axes. In a right-handed coordinate system, a positive circumpolar angle corresponds to a counter-clockwise angular displacement, when viewed from the positive pole. In a left-handed coordinate system, a positive circumpolar angle corresponds to a clockwise angular displacement, when viewed from the positive pole.

Note 1: The *handedness* of a coordinate system also indicates the positive direction assigned to the third axis, given the orientation of the polar and datum axes.

Note 2: These relationships are illustrated in **Figure 5-3 (Section 5.2.1)**.

3.23 IES Coordinate System A (Y, X)

A left-handed spherical coordinate system in a source

frame of reference, in which the pole is the third axis of the light source, while the direction of the receiver is given by an elevation angle (X) and a circumpolar angle (Y), measured from the first axis of the light source. (See **Figures 5-5** and **5-6**, **Section 5.4.1**.)

Note: This coordinate system is equivalent to the CIE (A , α) Coordinate System (see **Section 3.6**).

3.24 IES Coordinate System B (H , V)

A right-handed spherical coordinate system in a source frame of reference, in which the pole is the second axis of the light source, while the direction of the receiver is given by an elevation angle (H) and a circumpolar angle (V), measured from the first axis of the light source. (See **Figures 5-7** and **5-8**, **Section 5.4.2**.)

Note: This coordinate system is equivalent to the CIE (B , β) Coordinate System (see **Section 3.7**).

3.25 IES Coordinate System C (V , L)

A left-handed spherical coordinate system in a source frame of reference, in which the pole is aligned with a specified test axis for the light source, while the direction of the receiver is given by an apolar angle (V), and a circumpolar angle (L), measured from the half-plane containing the second axis of the light source. (See **Figures 5-9** and **5-10**, **Section 5.4.3**.) (See also **Section 3.33 test axis**.)

Note 1: In the typical case of zero tilt, the specified test axis is identical with the first axis of the light source, and the second axis of the light source is simply aligned with the instrument's datum axis ($L=0^\circ$, $V=90^\circ$). (See also **Section 3.35 tilt, measurement**.)

Note 2: This coordinate system is equivalent to the CIE (C , γ) Coordinate System (see **Section 3.8**).

3.26 left-handed coordinate system

(See **Section 3.22 handedness**.)

3.27 light source

The light-emitting device to be measured with a goniophotometer. This term includes lamps, solid state lighting products, luminaires, and any other type of light-emitting sources.

3.28 photometric center (of a light source)

The point in a light source under test that is specified, in the applicable test method or otherwise, as the point to be positioned at the center of rotation of the goniophotometer.

Note: The photometric center is typically defined in terms of the optical or mechanical properties of the source. For example, it may be defined as the centroid of a luminaire aperture, or as the estimated location of an equivalent point source, as calculated using the inverse-square law.

3.29 planar angle (angle in plane)

In a spherical coordinate system, an angle measured in a plane containing the pole. Such angular measurement may be made either from the positive polar direction (see **Section 3.1 polar angle**) or from the equatorial plane. (See also **Section 3.13 elevation angle** and **Figure 5-2** in **Section 5.1**.)

3.30 right-handed coordinate system

(See **Section 3.22 handedness**.)

3.31 second axis (of a light source)

An axis through the photometric center of a light source and perpendicular to the first axis.

Note: The first axis and second axis are defined by the manufacturer or by the photometric laboratory.

3.32 spherical coordinates

The coordinate systems used in photometry are variations on the standard spherical coordinate system used in mathematics. In the standard spherical coordinate system, a point in space is described by coordinates (ρ , θ , ϕ), as illustrated in **Figure 5-1** (**Section 5.1**). The first coordinate, rho (ρ), denotes a point's radial distance from the origin. The angular coordinates theta (θ) and phi (ϕ) together describe the direction to that point.

Note 1: The angle represented here by θ is described as an apolar angle, and is one type of planar angle. (See also **Section 3.13 elevation angle**.)

Note 2: The angle represented by ϕ is described as a circumpolar or azimuth angle.

Note 3: When describing points on a sphere, or direction only, the radius ρ is constant; therefore, this coordinate has little relevance in goniophotometry and is generally omitted.

3.33 test axis (of a source)

An axis through the photometric center of the light source in a direction specified by the photometric laboratory, or otherwise, and used to determine the orientation of the light source during goniophotometric measurement. (See also **Section 3.35 tilt, measurement.**)

Note: In the typical case of zero tilt, the specified test axis is identical with the first axis of the light source.

3.34 third axis (of a light source)

An axis through the photometric center of a light source and perpendicular to the first axis and the second axis.

3.35 tilt, measurement (of a source)

The directional offset between the first axis of a light source and the test axis specified for that source. In IES Coordinate System C, tilt direction may be specified in terms of the orientation of the first axis of the source within the test coordinate system.

Note 1: This term is not applicable to IES Coordinate Systems A or B because these coordinate systems are based directly on the first axis of the light source.

3.36 zenith angle

(See also **Section 3.1 apolar angle.**)

4.0 Laboratory, Goniometer, and Source frames of reference

4.1 Frames of Reference Applicable to Goniophotometry

A three-dimensional frame of reference can be represented by a set of three orthogonal axes, motionless with respect to a given object or observer, which may serve as the basis for the definition of a specific coordinate system. In the field of goniophotometry, at least four different frames of reference may be of interest: The source frame of reference, based on a

given light source under test, the instrument frame of reference, based on the design and function of the goniophotometer used to measure the source, the laboratory frame of reference, based on the space in which the testing is conducted, and the application frame of reference, based on the environment in which the light source is ultimately installed and operated. The standard coordinate systems considered in this document are all defined in terms of a *source frame of reference*.

4.1.1 Source Frame of Reference. The source frame of reference is defined by a set of three orthogonal axes intersecting at the specified photometric center of the light source, which serves as the origin of any associated coordinate system. Typically, the three axes are simply the first axis, second axis, and the third axis of the light source. The first axis of the light source passes through the photometric center, and generally corresponds to the direction in which the light source would be aimed. For luminaires, this is typically a line perpendicular to the aperture plane. For directional sources, it may be in the direction of maximum intensity, or the center of the source distribution, however defined. For non-directional sources, the definition of the first axis may be somewhat arbitrary, but is usually based on some prominent feature of the light source geometry. The second axis of the light source also passes through the photometric center and is perpendicular to the first axis. The choice of a second axis is generally based on some mechanical feature of the source, such as a long dimension, a mounting fixture, or a datum mark. The third axis of the light source also passes through the photometric center and is perpendicular to both the first and second axes. For this reason, once the first two axes are specified, the third axis is completely determined.

In some cases, it may be desirable to tilt the light source away from its specified design attitude during measurement. In such case, an alternative set of test axes may be defined by rotating the original set of axes in a specified direction. Of the IES standard coordinate systems, only IES Coordinate System C is defined in a way that allows for the specification of source tilt during measurement.

Figure 4-1 offers an abstract representation of a light source as a generic luminaire, with marking and shading added to help indicate luminaire orientation within the coordinate systems. The oval shaded region on one face of the luminaire represents the luminaire aperture. The first axis (i) of the luminaire is therefore perpendicular to this surface. The second axis (j) of this luminaire is here aligned with the long axis of the aperture. The third axis (k) is aligned with the short axis of the aperture. This set of axes is fixed with respect to the light source, regardless of any motion of the source relative to the laboratory, goniophotometer, or application environment. While **Figure 4-1** shows a left-handed light source frame of reference, the light source frame of reference may be left-handed or right-handed.

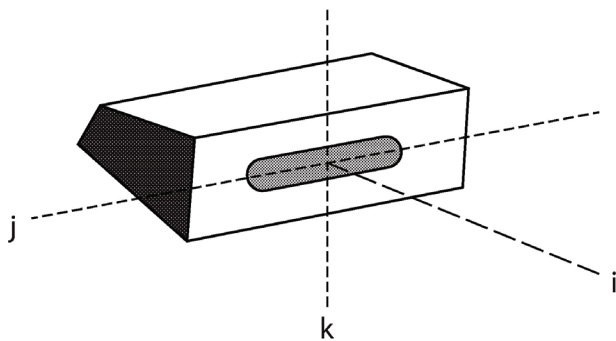


Figure 4-1. Light source frame of reference.

4.1.2 Goniometer Frame of Reference. The goniometer frame of reference is defined with respect to the optical axis of detection and the axes of rotation of a given goniophotometer system. The origin of this frame of reference is the goniometer's center of rotation—that is, the intersection of its two axes of rotation—which normally coincides with the photometric center of the light source during measurement. In this frame of reference, both the source and the laboratory rotate during measurement, while the axis of detection remains fixed.

4.1.3 Laboratory Frame of Reference. A laboratory frame of reference is defined with respect to the architectural elements of the environment in which a light source is tested. The most straightforward origin for such a frame of reference is the goniometer's center of rotation, which normally coincides with the photometric center of the light source during measurement.

Depending on the goniophotometer design, both the light source and the detector may move during measurement of the source, while the position of the laboratory (i.e., its floor, walls, and ceiling) remains fixed. This is the frame of reference in which the goniometer system is naturally represented during the original setup and alignment of the instrument, and the positioning and orientation of the light source under test. It is also the most straightforward frame of reference within which the stray-light performance of the system may be characterized, since the distribution of stray light is related to reflection from various features of the laboratory environment, such as walls, floor, ceiling, doors, thermal vents, and auxiliary lighting systems.

4.1.4 Application Frame of Reference. The application frame of reference is defined with respect to the field environment in which a light source is installed and operated. If, upon installation, the source axes are aligned with those of the application environment, the origin of this frame of reference coincides with the photometric center of the light source, and the application frame of reference is equivalent to the source frame of reference.

4.2 Orientation with Respect to Gravity

In the literature of goniophotometry, terms such as *vertical*, *horizontal*, *zenith*, and *nadir* are used in various, sometimes inconsistent and misleading, ways. The terms *vertical* and *horizontal* are used in some cases to indicate the physical orientation of an object with respect to the local gravitational vector, or “true vertical”; in other cases, they are used simply to make a distinction between two mutually perpendicular, but otherwise arbitrarily oriented features. In some cases, the terms *zenith* and *nadir* are used in a purely geometric sense, in which *zenith* refers to the positive polar direction of a spherical coordinate system, and *nadir* refers to the negative polar direction. In other cases, these terms are used to indicate physical orientation with respect to gravity.

The influence of gravity on the operation of a light source and the orientation of a light source with respect to “true vertical” are important considerations, and these correspond to certain specific requirements regarding goniophotometer design and test conditions, as described in **Section 6**. When it comes to the definition

4. Choose the direction of positive circumpolar rotation: the *handedness* of the system.
5. Choose a type of *planar angle*: either an *apolar angle*, measured from the positive polar direction, or an *elevation angle*, measured from the equatorial plane.
6. Specify the *coordinate scale* in units of angle (degrees or radians).

Handedness: The *handedness* of a spherical coordinate system describes the direction of positive circumpolar rotation, in relation to the orientation of the system's polar and datum axes. In a *right-handed* coordinate system, a positive circumpolar angle corresponds to a counterclockwise angular displacement, when viewed from the positive pole. In a *left-handed* coordinate system, a positive circumpolar angle corresponds to a clockwise angular displacement, when viewed from the positive pole. It is important to note that by trigonometric convention, the handedness of a coordinate system also indicates the positive direction assigned to the system's tertiary axis, given the orientation of the polar and datum axes. These definitions of *handedness* correspond to the following mnemonic procedures, illustrated in **Figure 5-3** for a right-handed system:

- For a right-handed system, point your right thumb in the positive polar direction (+p). Your fingers will curl in the direction of increasing circumpolar angle.
- For a right-handed system, point your right thumb in the positive polar direction (+p), and your extended index finger in the direction of the datum axis (+d). Your middle finger, extended at right angles to your index, will point in the positive direction of the tertiary axis (+t).

- For a left-handed system, point your left thumb in the positive polar direction (+p). Your fingers will curl in the direction of increasing circumpolar angle.
- For a left-handed system, point your left thumb in the positive polar direction (+p), and your extended index finger in the direction of the datum axis (+d). Your middle finger, extended at right angles to your index, will point in the positive direction of the tertiary axis (+t).

5.3 Example: The Geographic Coordinate System

The standard Geographic Coordinate System, which uses the coordinates of *latitude* and *longitude* to identify locations on Earth, is like the spherical coordinate systems considered here. Longitude is a circumpolar angle, and defines locations from 180° west through 0° to 180° east. Latitude is an angle of elevation, and defines locations from 90° south through 0° (the equator) to 90° north. The map of the earth shown in **Figure 5-4** marks several locations: In geographic coordinates, Greenwich, England is located at (51°N, 0°E). The South American city of Quito, Ecuador has a geographic location of (0°S, 78°W). On the remote Pacific isle of Fiji, the town of Suva is situated at (16°S, 180°E). The geographic origin (0°N, 0°E) lies off the African coast, deep in the waters of the Gulf of Guinea. On the frozen continent of Antarctica, the geographic South Pole is designated as (90°S, 0°E).

It should be noted that if northern latitudes and western longitudes were assigned a positive value, while southern latitudes and eastern longitudes were assigned a negative value, then this system would be recognizable as a left-handed spherical coordinate system. To clarify

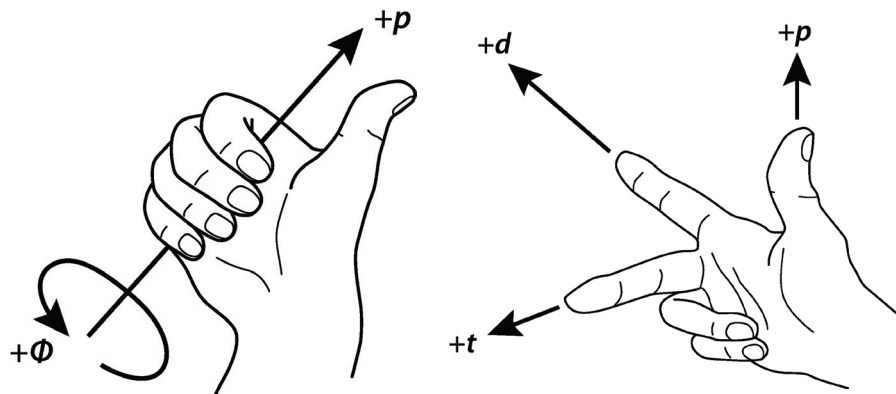


Figure 5-3. Handedness of a spherical polar coordinate system.

of coordinate systems, however, it should be recognized that each of the standard goniophotometric coordinate systems described in this document is defined in a source frame of reference, independent of the direction of gravity. To avoid confusion, the term *zenith* and its antonym *nadir* have generally been avoided here. The terms *vertical* and *horizontal*, however, are too well established as names for various angular coordinates to be avoided entirely.

5.0 Coordinate Systems

5.1 Spherical Coordinate Systems

The coordinate systems used in goniophotometry are variations on the standard spherical coordinate system used in mathematics. In the standard system, a point in space is specified by coordinates (ρ, θ, ϕ) , as illustrated in **Figure 5-1**. The first coordinate, *rho* (ρ), denotes the point's radial distance from the origin, while the angular coordinates *theta* (θ) and *phi* (ϕ) together describe the direction to that point.

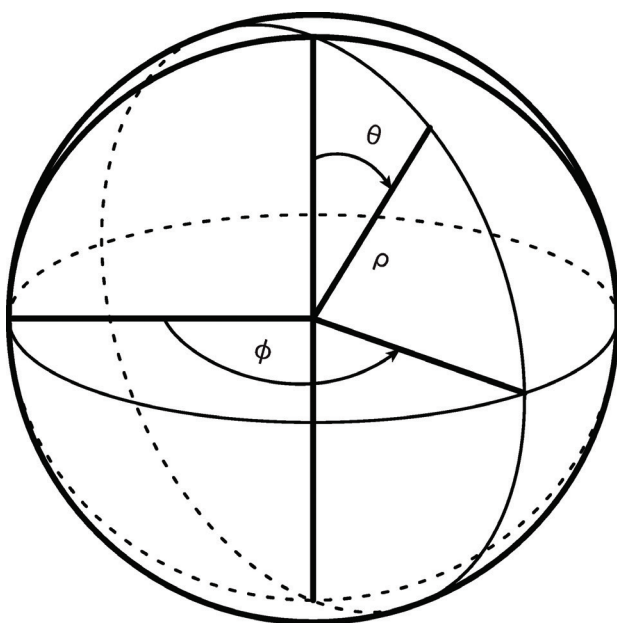


Figure 5-1. Standard spherical coordinate system specified by coordinates (ρ, θ, ϕ) .

As illustrated in **Figure 5-2**, the angle represented by θ may be described as the *apolar angle*, and denotes

the angular displacement of the point from the positive polar direction. The apolar angle is one of two types of angles commonly used to measure direction within a polar plane. Some coordinate systems use instead the *elevation angle*, which may be denoted as ϵ . The angle represented by ϕ is described as the *circumpolar angle*, and denotes the angular displacement of the polar plane that contains the point from the *datum plane*, which contains the system's *datum axis*. When describing points on the surface of a sphere, ρ is constant; when describing direction only, ρ is irrelevant. For this reason, in goniophotometry, a two-dimensional, *abridged spherical coordinate system* is used, and is denoted by angular coordinates (θ, ϕ) alone.

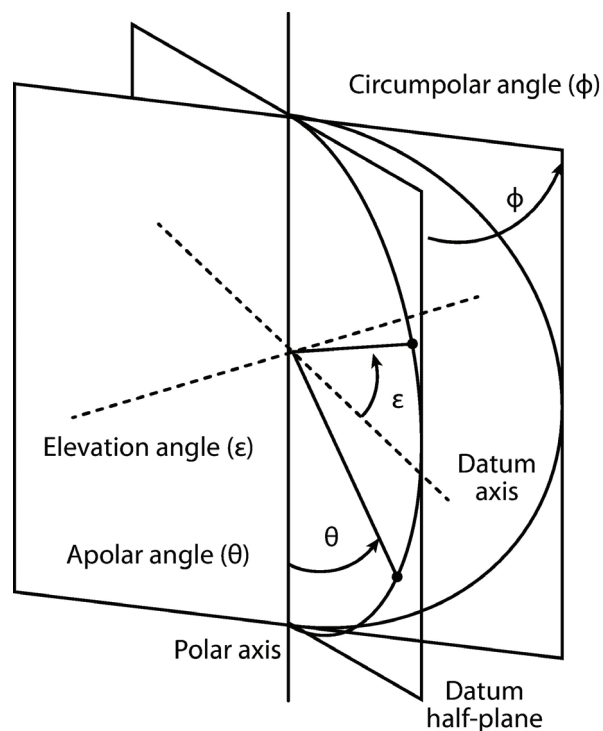


Figure 5-2. Angles from and within half-planes.

5.2 Defining a Spherical Coordinate System

The standard coordinate systems presented in **Section 5.4** are abridged spherical coordinate systems, which have each been defined using the following steps:

1. Specify a three-dimensional *frame of reference*, comprising three orthogonal axes.
2. Select a *polar axis*, and assign its positive polar direction (+).
3. Select a *datum axis*, and assign its positive polar direction (+).

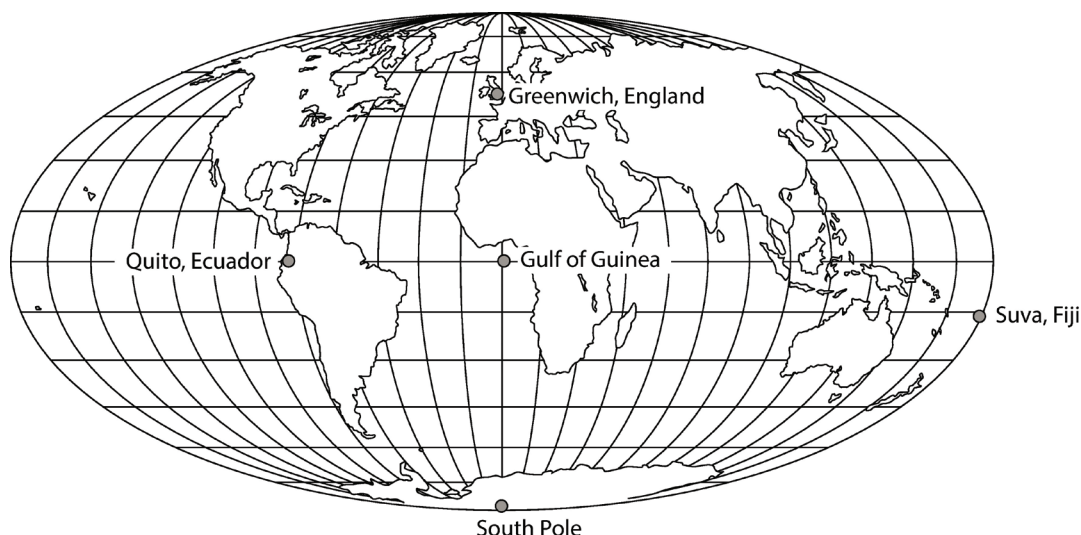


Figure 5-4. The Standard Geographic Coordinate System

the explanation of the goniophotometric coordinate systems described below, examples in the following text relate back to the coordinates of the specific geographic locations identified in **Figure 5-4**.

5.4 Standard Goniophotometric Coordinate Systems

5.4.1 IES Coordinate System A (IES-A). Automotive lighting equipment and optical systems are typically tested and reported using IES Coordinate System A (IES-A) (see **Figure 5-5**). The polar axis ($+p$) is represented as vertical,* and angles measured in the polar planes are described as vertical angles (Y). The light source is positioned such that its first axis lies on the equator ($0^\circ Y$), and serves as the datum axis ($+d$). Circumpolar angles from the datum axis to the measurement half-plane are described as horizontal angles (X). Measurement direction is denoted by the corresponding (Y, X) coordinate pair.

IES-A is a left-handed system: the direction of increasing horizontal angle is clockwise when viewed from the positive polar direction ($90^\circ Y$). Vertical angles (Y) range from -90° to 90° , and horizontal angles (X) range from -180° to 180° .

* It is important to note that although the polar axis is represented as vertical, and the apolar angle in this system is designated as the “vertical angle,” these designations are arbitrary and are not necessarily related to the direction of gravity.

IES Coordinate System A could be used to identify the geographical locations represented in **Figure 5-4** as follows: Greenwich, England would be located at ($51^\circ Y, 0^\circ X$), Quito, Ecuador at ($0^\circ Y, 78^\circ X$), and Suva, Fiji at ($-16^\circ Y, 180^\circ X$). The point at ($0^\circ Y, 0^\circ X$) would lie in the Gulf of Guinea, while the location of the geographic South Pole would be ($-90^\circ Y, 0^\circ X$).

IES Coordinate System A is equivalent to the CIE (A, α) Coordinate System, defined in CIE 121-1996 and illustrated in **Figure 5-6**.³

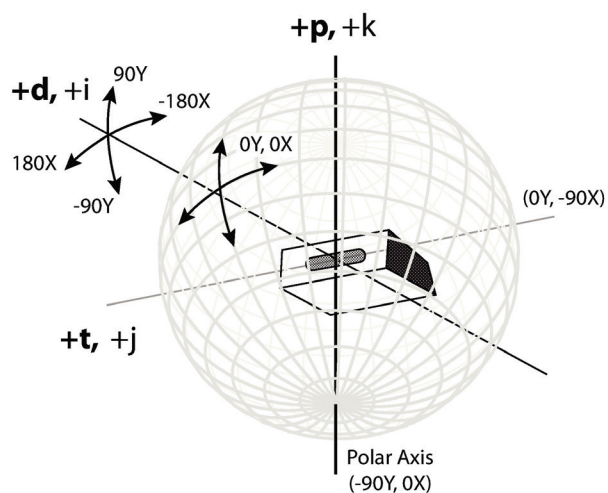


Figure 5-5. IES Coordinate System A (IES-A); arrows indicate photometric angles.

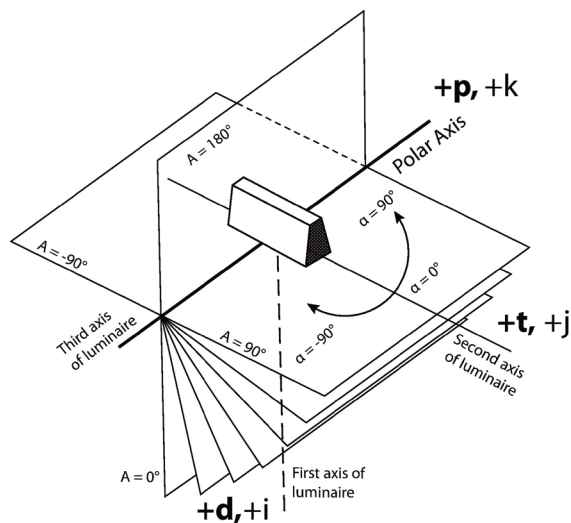


Figure 5-6. The CIE (A, α) Coordinate System, as described in CIE 121-1996.

5.4.2 IES Coordinate System B (IES-B). Floodlight and searchlight photometric data are traditionally presented in IES Coordinate System B (IES-B), which is illustrated in **Figure 5-7**. It should be noted that IES-B looks much like the IES-A coordinate system, turned on its side. The polar axis (+p) is represented as horizontal,* and angles measured in the polar planes are described as horizontal angles (H). The first axis of the light source lies on the equator ($0^\circ H$), and serves as the datum axis (+d). Circumpolar angles from the datum axis to the measurement half-plane are described as vertical angles (V). Measurement direction is denoted by the corresponding (H, V) coordinate pair.

IES-B is a right-handed system: the direction of increasing vertical angle is counterclockwise when viewed from the positive polar direction ($90^\circ H$). Horizontal angles (H) range from -90° to 90° , and vertical angles (V) range from -180° to 180° . Data are commonly provided only from $-90^\circ V$ to $+90^\circ V$, however, because the sources under test are normally directional.

IES Coordinate System B could be used to identify the geographical locations represented in **Figure 5-4** as

* It is important to note that although the polar axis is represented as horizontal, and the apolar angle in this system is designated as the "horizontal angle," these designations are arbitrary, and are not necessarily related to the direction of gravity.

follows: Greenwich would be at ($0^\circ H, 51^\circ V$), Quito at ($78^\circ H, 0^\circ V$), and Suva, Fiji, at ($0^\circ H, -164^\circ V$). The point at ($0^\circ H, 0^\circ V$) would be in the Gulf of Guinea, while the South Pole would be designated by ($0^\circ H, -90^\circ V$).

IES Coordinate System B is equivalent to the CIE (B, β) Coordinate System, defined in CIE 121-1996 and illustrated in **Figure 5-8**.³

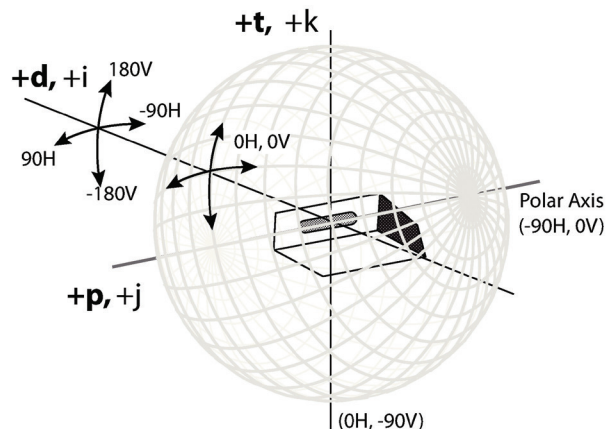


Figure 5-7. IES Coordinate System B (IES-B): arrows indicate photometric angles.

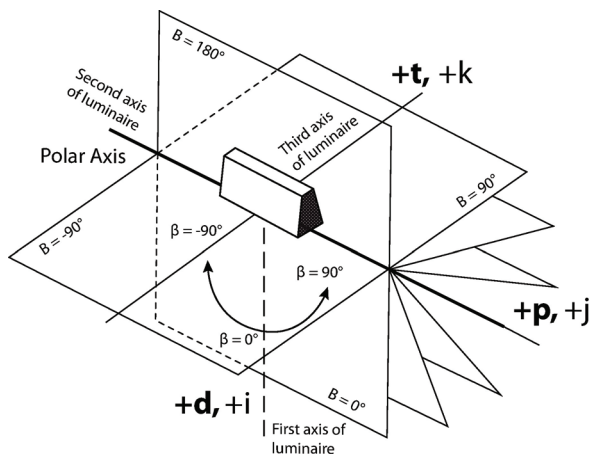


Figure 5-8. The CIE (B, β) Coordinate System, as represented in CIE 121-1996.³

5.4.3 IES Coordinate System C (IES-C). IES Coordinate System C (IES-C) is the most commonly used and widely recognized goniophotometric coordinate system. Most goniophotometric data, including data for indoor and roadway luminaires, are acquired and presented using this coordinate system, which is illustrated in **Figure 5-9**. The polar axis (+p) is represented as vertical, and angles

measured in the polar planes are described as vertical angles (V). Circumpolar angles are described as lateral angles (L). Measurement direction is denoted by the corresponding (V, L) coordinate pair.

IES-C is a left-handed system: the direction of increasing lateral angle is clockwise, when viewed from the positive polar direction ($0^\circ V$). Vertical angles (V) range from 0° to 180° . Lateral angles (L) range from 0° to 360° . The first axis of the light source is typically aligned with the polar axis, but the light source may instead be tilted away from the IES-C axis in a specified direction. In the case of zero tilt, the first axis of the light source serves as the polar axis of the coordinate system, and the second axis of the source serves as the datum axis, as shown in **Figure 5-9**.

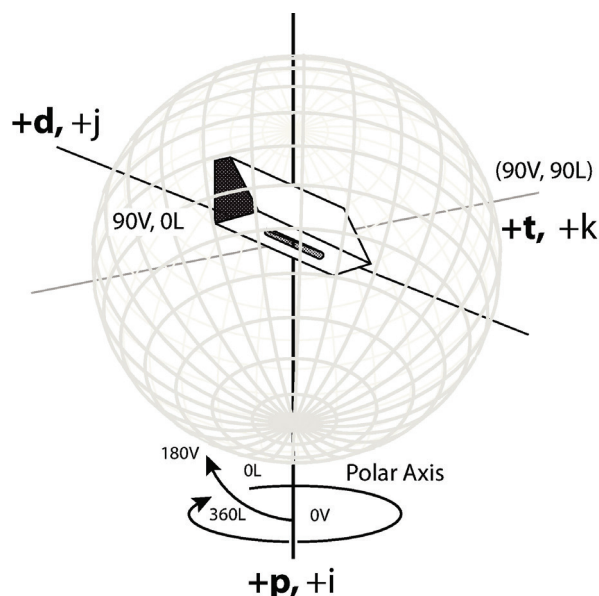


Figure 5-9. IES Coordinate System C (IES-C): arrows indicate photometric angles.

IES Coordinate System C could be used to identify the geographical locations represented in **Figure 5-4** as follows: Greenwich would be at ($141^\circ V$, $0^\circ L$), Quito at ($90^\circ V$, $282^\circ L$), and Suva, Fiji at ($74^\circ V$, $180^\circ L$). The point at ($90^\circ V$, $0^\circ L$) would lie in the Gulf of Guinea, and the South Pole would have coordinates ($0^\circ V$, $0^\circ L$).

IES Coordinate System C is equivalent to the CIE (C, γ) Coordinate System, defined in CIE 121-1996 and illustrated in **Figure 5-10**.³

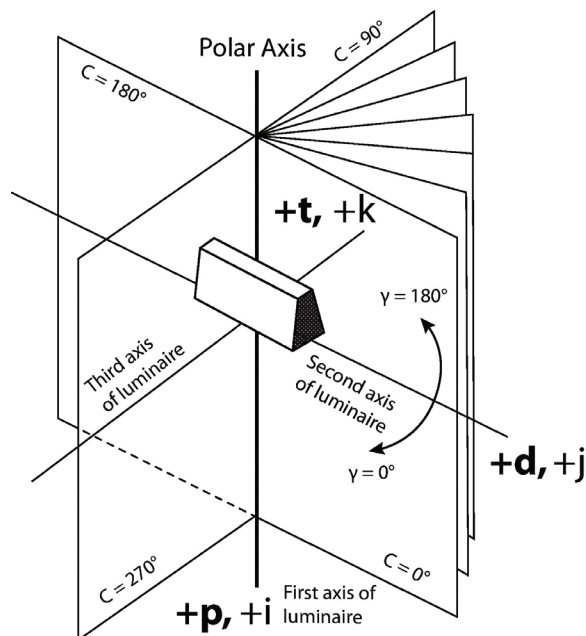


Figure 5-10. The CIE (C, γ) Coordinate System, as represented in CIE 121-1996.³

6.0 Goniophotometers

6.1 Type A Goniophotometer

6.1.1 Instrument Design and Setup. A Type A goniophotometer system is designed with a polar axis and secondary axis, which physically correspond to the axes of the IES-A coordinate system (see **Figure 5-5, Section 5.4.1**).

In a Type A goniophotometer, the photodetector position is fixed, while the light source is rotated about the instrument's polar axis and secondary axis, as shown in **Figures 6-1** and **6-2**.

To correlate the goniophotometer coordinates directly with those of the IES-A Coordinate System, the third axis (k) of the light source is aligned with the instrument's polar axis, and its first axis (i) is aligned with the instrument's datum axis, such that the photodetector lies in the $+i$ direction when the goniophotometer is positioned at coordinates ($0Y, 0X$).

6.1.2 Instrument Operation. The light source is first rotated about the polar axis to select a data plane

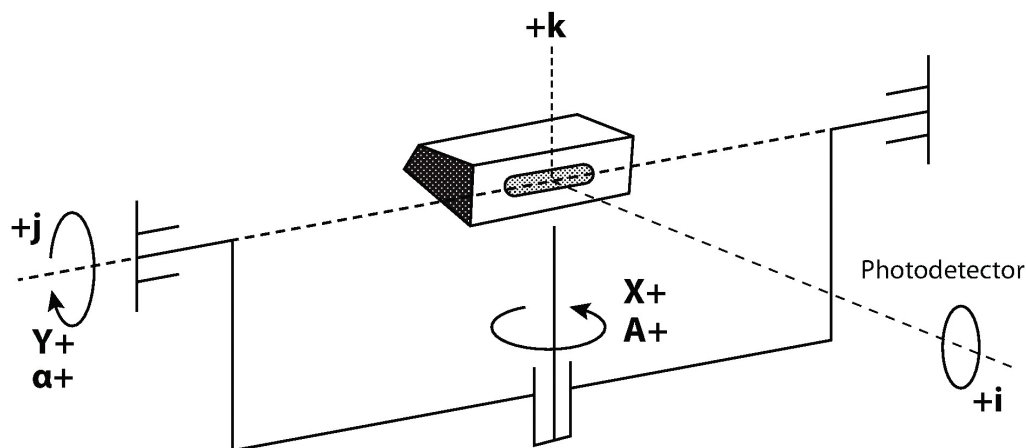


Figure 6-1. Schematic representation of a Type A Goniophotometer. Arrows indicate the direction of positive rotation about each axis.

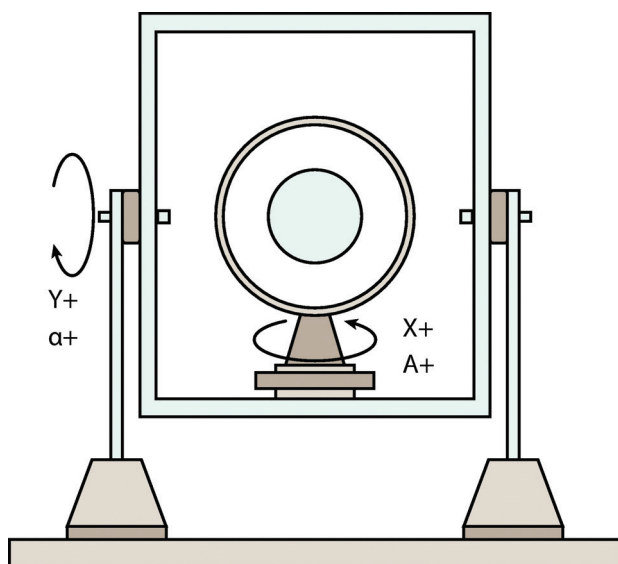


Figure 6-2. Alternative representation of a Type A goniophotometer. The detector is in the direction of the viewer of the figure. Arrows indicate the direction of positive rotation about each axis.

designated by the instrument's X ordinate. The source is then rotated through the full range of Y ordinates, until a full plane of data has been acquired. This process is repeated for a full set of data planes (X) until the light source has been measured at all desired (Y, X) coordinates.

6.1.3 Applications. Type A goniophotometers are widely used for automotive device photometry, and may be used for other applications. It is important to note, however, that because the Type A goniophotometer

relies on tilting the light source with respect to gravity to make measurements, it is not suitable for use with position-sensitive light sources.

6.2 Type B Goniophotometer

6.2.1 Instrument Design and Setup. A Type B goniophotometer system is designed with a polar axis and secondary axis, which physically correspond to the axes of the IES-B coordinate system (see **Figure 5-7, Section 5.4.2**).

In a Type B goniophotometer, the photodetector position is fixed, while the light source is rotated about the instrument's polar axis and secondary axis, as shown in in **Figures 6-3** and **6-4**.

To correlate the goniophotometer coordinates directly with those of the IES-B Coordinate System, the second axis (j) of the light source is aligned with the instrument's polar axis, and its first axis (i) is aligned with the instrument's datum axis, such that the photodetector lies in the $+i$ direction when the goniophotometer is positioned at coordinates ($0H, 0V$).

6.2.2 Instrument Operation. The light source is first rotated about the polar axis to select a data plane designated by the instrument's H ordinate. The source is then rotated through the full range of V ordinates, until a full plane of data has been acquired. This process is repeated for a full set of data planes (H) until the light source has been measured at all desired (H, V) coordinates.

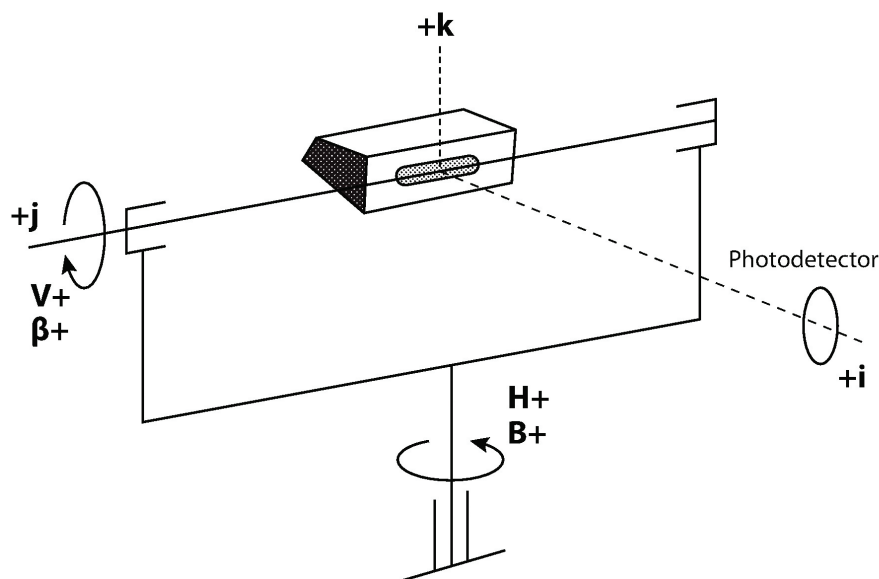


Figure 6-3. Schematic representation of a Type B goniophotometer. Arrows indicate the direction of positive rotation about each axis.

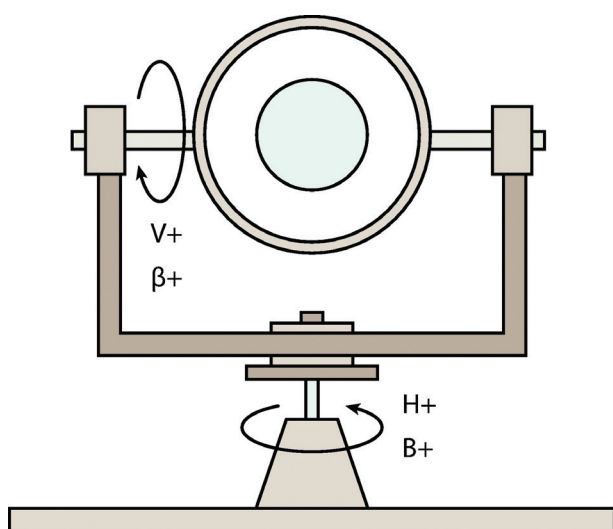


Figure 6-4. Alternative representation of a Type B goniophotometer. The detector is in the direction of the viewer of the figure. Arrows indicate the direction of positive rotation about each axis.

6.2.3 Applications. Type B goniophotometers may be used for various applications. It is important to note, however, that because the Type B goniophotometer relies on tilting the light source with respect to gravity to make measurements, it is not suitable for use with position-sensitive light sources.

6.3 Type C Goniophotometer

6.3.1 Instrument Design and Setup. A Type C goniophotometer system is designed with a polar axis and secondary axis, which physically correspond to the axes of the IES-C coordinate system (see **Figure 5-9, Section 5.4.3**).

In a Type C goniophotometer, the light source is mounted in a fixed orientation with respect to gravity, rotating about the instrument's vertically-oriented polar axis, while the photodetector (or virtual photodetector) rotates with respect to the light source about the goniophotometer's secondary axis, as shown in in **Figure 6-5**.

To correlate the goniophotometer coordinates directly with those of the IES-C coordinate System, the first axis (*i*) of the light source is aligned with the instrument's polar axis, such that the direction of detection coincides with the *+i* direction with the goniophotometer positioned at $L = 0$. The second axis (*j*) of the light source is aligned with the instrument's datum axis, such that the direction of detection coincides with the *+j* direction with the goniophotometer positioned at coordinates $(90V, 0L)$.

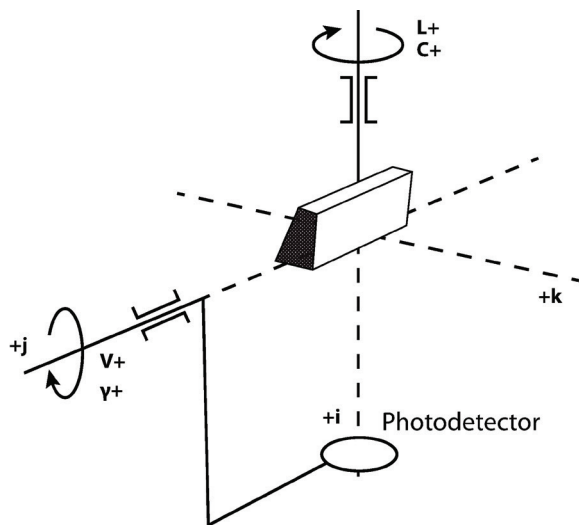


Figure 6-5. Schematic representation of a Type C goniometer. Arrows indicate direction of positive luminaire rotation. The direction of positive rotation of the light source about the L axis, when viewed from the positive polar direction, is counterclockwise. Conversely, in the source frame of reference, the direction of positive rotation of the detector, when viewed from the positive pole, is clockwise.

6.3.2 Instrument Operation. The light source is first rotated about the polar axis to select a data plane designated by the instrument's L ordinate. The direction of detection is then determined by rotation of the photodetector through the full range of V ordinates, until a full plane of data has been acquired. This process is repeated for a full set of data planes (L) until the light source has been measured at all desired (V, L) coordinates.

6.3.3 Applications. The constant orientation of the light source with respect to gravity makes the Type C goniophotometer ideal for measuring the light output of position-sensitive light sources. Thus, it is commonly used in the photometry of interior and street light sources.

A Type C goniophotometer can also be configured in such a way as to correlate its angular coordinates with those of the IES-A Coordinate System (see **Figure 5-5, Section 5.4.1**). In this case, the measurement procedure is like that outlined above, except that the light source is initially positioned so that its third axis (k) is aligned with the instrument's polar axis, and its first axis (i) is aligned with the instrument's datum axis. The data acquired in this fashion will correspond point-for-point with the

grid of points acquired by a Type A goniophotometer. The angular coordinates of the Type C instrument can be mapped onto the corresponding IES-A coordinates by a relatively simple transformation (see **Annex A**).

6.3.4 Design Variations. The essential function of the Type C goniophotometer can be realized in various ways: The photodetector may literally rotate about the light source, as illustrated schematically in **Figure 6-5**, or a virtual photodetector may be rotated about the source by means of a moving mirror, while the detector position remains fixed. Furthermore, the rotation of the photodetector with respect to the light source may be accomplished by moving the photodetector or mirror alone, or by moving both the source and photodetector or mirror about a common axis. It is important to note, however, that in all Type C goniophotometers, the orientation of the light source shall be fixed with respect to gravity. Examples of different Type C goniophotometer designs are described in the following sections and illustrated in **Figures 6-6** and **6-7**.

Moving detector goniophotometers: In this type of goniophotometer, a photodetector moves on a rotating boom or arc-shaped track, with the light source positioned at the center of detector rotation. Readings are collected with the detector positioned at the desired angular settings. In an alternative version of this approach, multiple detectors, positioned in an arc, replace the moving detector. **Figure 6-6** shows examples of moving detector goniophotometers.

Moving mirror goniophotometer: In this type of goniophotometer, a mirror is rotated with respect to the light source, reflecting the luminous flux to a single, fixed detector. Readings are taken at each desired angle as the mirror moves to that location.

Figure 6-7 shows two examples of moving mirror goniophotometers. A third variation exists, in which the source and mirror each rotate about a common axis (compare **Figure 6-6[c]**). For variations that require the luminaire to move through a circular pattern while the goniophotometer is running, it is important to consider carefully the speed at which the source is moving, so as not to disturb the arc in discharge-based sources, and to prevent changes in convective cooling, which could affect the accuracy of the results.

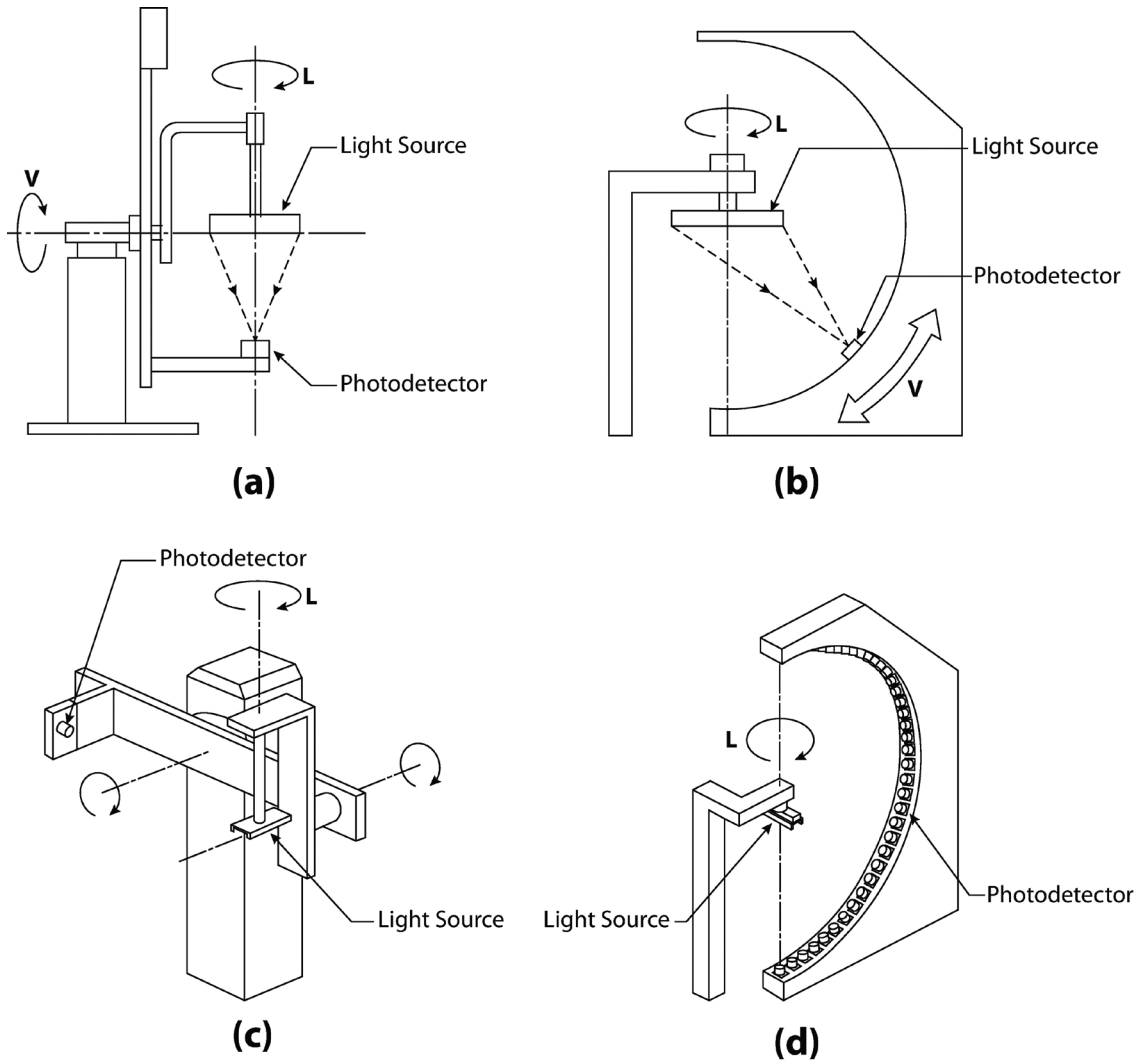


Figure 6-6. Examples of moving detector goniophotometers.

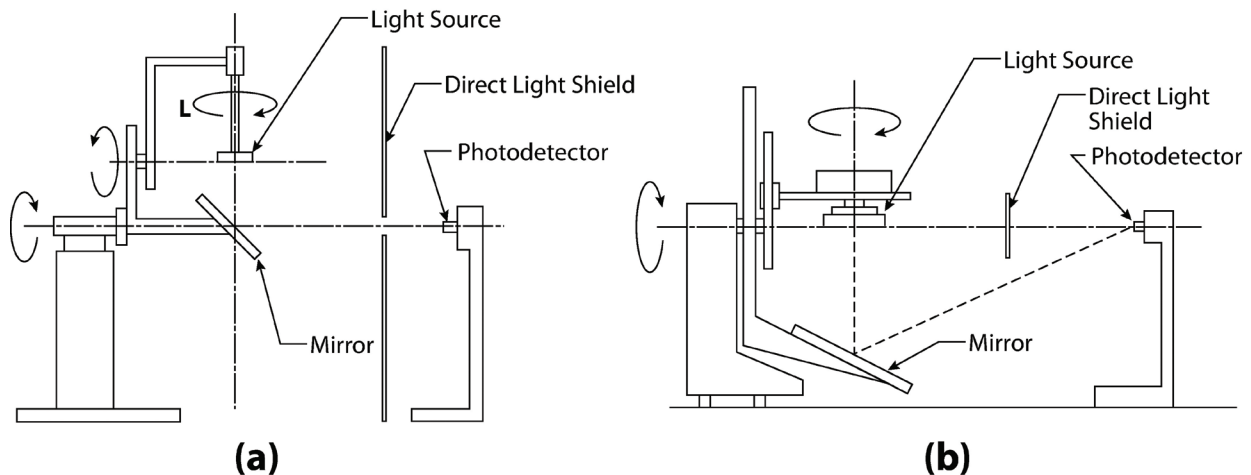


Figure 6-7. Examples of moving mirror goniophotometers.

6.4 Type D Goniophotometer

6.4.1 Instrument Design and Setup. Like a Type C goniophotometer, a Type D goniophotometer system is designed with a polar axis and secondary axis, which physically correspond to the axes of the IES-C Coordinate System (see **Figure 5-9, Section 5.4.3**). In a Type D goniophotometer, the photodetector position is fixed, while the light source is rotated about the instrument's polar axis, which is typically horizontally oriented, and its secondary axis, as shown in in **Figure 6-8**. To correlate the goniophotometer coordinates directly with those of the IES-C Coordinate System, the first axis (i) of the light source is aligned with the instrument's polar axis, and its second axis (j) is aligned with the instrument's datum axis, such that the direction of detection coincides with the $+j$ direction with the goniophotometer positioned at coordinates $(90^\circ, 0^\circ)$.

6.4.2 Instrument Operation. The light source is first rotated about the polar axis to select a half-plane designated by the instrument's L ordinate. The source is then rotated through the full range of V ordinates (0° to 180°), until a half-plane of data has been acquired. This process is repeated for the full range of L ordinates (0° to 360°), until the light source has been measured at all desired (V, L) coordinates.*

* An alternative procedure is to measure a full plane of data (V ranging from -180 to $+180$) for each L position. This need only be repeated over half the range of L ordinates (0° to 180°) in order to span the complete sphere.

6.4.3 Applications. As with the Type A and B goniophotometers, the Type D goniophotometer tilts the light source to take measurements; therefore, it is not suitable for use with position-sensitive light sources.

7.0 Calibration

There are two basic methods to calibrate goniophotometric test data, relative photometry and absolute photometry. Each method has advantages and disadvantages.

7.1 Relative Photometry Method

In relative photometry, the goniophotometer is used to measure the luminaire efficiency independent of the test lamp. The data from this bare lamp test are integrated to create a value typically referred to as the bare lamp sum. The *bare lamp sum* is a number proportional to the lumen output of the lamp. This bare lamp sum and the integrated sum of the luminaire intensity data are then used to normalize the luminous intensity data of the luminaire. The normalized luminaire data can then be rescaled by the manufacturer's initial rated lumen output for the test lamp to create report data or IES file data that are prorated for the lamp's rated lumen output. From a practical standpoint, the relative photometry method

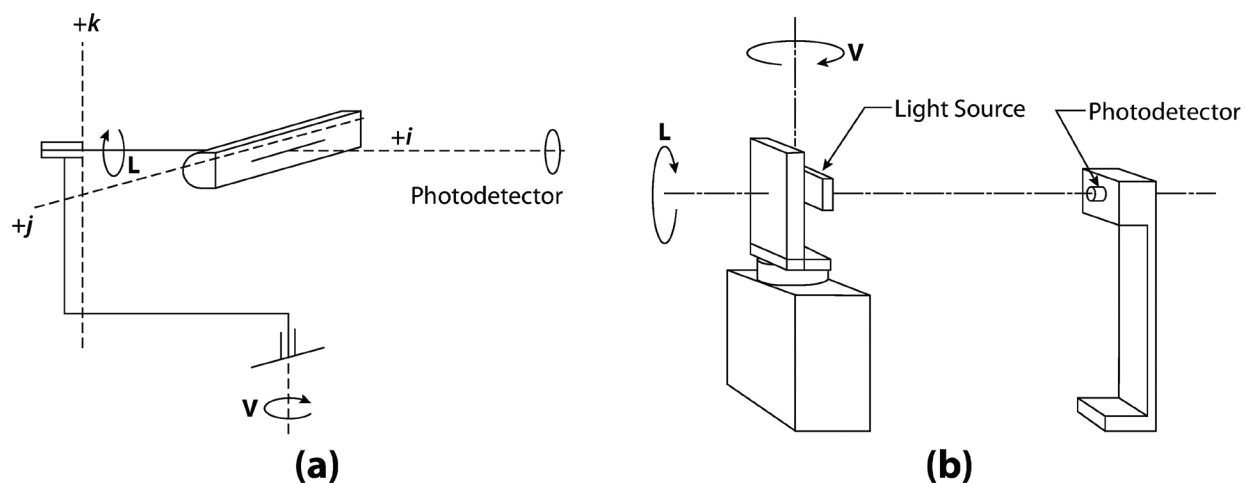


Figure 6-8. Schematic diagram (a) and example (b) of a Type D goniophotometer. Arrows indicate direction of light source rotation.

is desirable because a traceable photometric standard is not necessary.

It is important to note that the applicability of relative photometry is highly dependent on the consistent operation of the test lamps between the luminaire and bare lamp measurements. Therefore, it is important that both tests use the same control gear operating the same lamps in the same tilt position. It is also recommended that the same power supply and electrical measurement instruments be used. If possible, the driver and lamp holders from the luminaire should be removed and installed in a bare lamp fixture for determination of the bare lamp sum. The driver and lamp holders are then reinstalled in the test luminaire. To take advantage of the benefits of the relative photometric procedure, photometric measurements shall be acquired using the same goniophotometric system.

In practice, relative photometry is typically performed according to the following steps:

- Measure the bare lamp(s). The bare lamp is measured in the same tilt position, under the same electrical operating conditions, and on the same goniophotometric system as will be used for the luminaire test.
- Measure the luminaire. The luminaire is measured using the same test lamps, under the same electrical operating conditions, and on the same goniophotometric system as was used for the bare lamp test.
- Process the luminous intensity data:
 - Determine the ratio of the bare lamp test signal gain to the luminaire test signal gain.
 - Determine the manufacturer's initial lamp lumen rating for the test lamp.
 - Compute the test lamp's bare lamp sum by summing the product of the measured intensity data and the zonal constants for the measurement angle set. The zonal constant calculation is described in **Annex C**.
 - Scale the measured intensity data according to the equation below to create 'corrected' luminous intensity values that are prorated to the test lamp's rated initial lumen output.

$$I_c = \frac{I_m \times LLR}{BLS} \times R_G,$$

where:

I_c = Corrected luminous intensity value of luminaire at a given angle

I_m = Measured luminous intensity value of luminaire at a given angle

BLS = Bare lamp sum

LLR = Lamp lumen rating

R_G = Ratio of measurement gains for the bare lamp and luminaire tests

- Compute the luminaire efficiency:
- Compute the *luminaire sum* by summing the product of the measured luminaire intensity data and the zonal constants for the luminaire's measurement angle set:

$$LS = \iint_{\theta, \phi} I_{c(\theta, \phi)} \times K_{(\theta_1 - \theta_2, \phi_1 - \phi_2)},$$

where:

LS = Luminaire sum

I_c = Corrected luminous intensity value of luminaire at a given angle

K = Zonal Constant (as described in **Annex B**) from a range of apolar and circumpolar angles (θ_1 through θ_2 , and ϕ_1 through ϕ_2)

- Compute the luminaire efficiency:

$$E_L = \frac{LS}{BLS} \times R_G,$$

where:

EL = Luminaire efficiency

LS = Luminaire sum

BLS = Bare lamp sum

RG = Ratio of measurement gains for the bare lamp and luminaire tests

The order in which the bare lamps and luminaire are tested is not important; however, the bare lamps shall be measured immediately before or after the luminaire test. This will ensure that any changes to the photometric system or lumen depreciation of the test lamps will have a minimum impact on the results.

7.2 Absolute Photometry Method

In absolute photometry, the goniophotometer is calibrated by establishing the relationship between the raw signal,

or uncorrected reading (indication) of the photometer, and the corresponding luminous intensity. This may be accomplished either by using a mathematical model based on system component data (e.g., mirror reflectivity or photodetector responsivity) or by measuring the response of the system as a whole to a traceable standard of photometric or radiometric quantities. The calibrated goniophotometer can then be used to measure the properties of a light source or a luminaire in terms of absolute luminous intensity. Absolute photometry is advantageous when light source properties are strongly influenced by their use in the luminaire due to thermal effects such that independent source measurements are not representative of the luminaire output. In addition, the absolute photometry method is necessary for luminaires whose light sources cannot be separated from the luminaire. The disadvantages of absolute photometry are that a traceable standard of photometric or radiometric quantities needs to be maintained, and that the absolute photometry method alone cannot measure luminaire efficiency in terms of light loss due to design.

There are three calibration methods for goniophotometers: luminous intensity calibration, illuminance calibration, and total luminous flux calibration. (Similarly, goniospectroradiometers are calibrated using the radiometric equivalents: spectral radiant intensity calibration, spectral irradiance calibration, and total spectral radiant flux calibration.) The suitable calibration method should be chosen based on the goniophotometer type and the purpose of the measurement.

Absolute calibration methods involve the use of standard devices (lamps or detectors) that are traceable to SI units* through photometric standards provided by a national metrology institute (NMI)†. The standard devices may include primary standards, working standards, and/or transfer

standards. The standard devices should be recalibrated at an interval specified in terms of total operating time, which depends on the aging rate of devices. For standard lamps, it is recommended that a set of three or more standard lamps be maintained as primary standard lamps for a laboratory, and that their scale be transferred to working standard lamps for routine calibration work, to reduce the operating time of the primary standard lamps. The similar use of a set of working standard detectors is also recommended for primary standard detectors.

Standard lamps should be operated according to specifications and instructions provided by the calibration laboratory. Lamp current or voltage is set as specified. A standard lamp shall be ramped up and down slowly (typically, 30 s) to avoid high inrush currents and associated localized thermal shock to the filaments. Photometric measurements shall be made after the lamp has stabilized, which normally takes five to ten minutes. Lamp current, voltage, and light output may be monitored to determine when a specified criterion of stability has been achieved. The lamp should not be operated longer than necessary, to keep the total operating time of the lamp to a minimum. A record of the operating time of each standard lamp should be maintained.

7.2.1 Luminous Intensity Calibration

7.2.1.1 Luminous Intensity Standard. A luminous intensity standard is a light source that has been calibrated by a laboratory capable of establishing an absolute value of luminous intensity (measured in candelas), such as an NMI or commercial calibration laboratory. Testing laboratories may also calibrate their own instrumentation using internally calibrated working standard lamps whose luminous intensity is transferred from SI-traceable standards.

For goniophotometer calibration, the standard lamp should have nearly uniform luminous intensity distribution around the direction of calibration. Since the photometric distance at which the standard lamp is calibrated is usually much shorter than the distance at which it is measured by the goniophotometer, any non-uniformity of luminous intensity distribution may cause an error in the calibration. Conversely, a uniform distribution serves to minimize such error. Uniform luminous intensity distribution can

* SI units: The International System of Units. A system of physical units based on the meter, kilogram, second, ampere, kelvin, candela, and mole, together with a set of prefixes to indicate multiplication or division by a power of ten. (Google Dictionary. International System of Units; [accessed 2018 Jan 6] www.google.com/search?q=Dictionary#dobs=SI%20units).

† A national metrology institute's (NMI) role in a country's measurement system is to conduct scientific metrology, realize base units, and maintain primary national standards. (Metrology. Wikipedia; [accessed 2018 Jan 6] <http://en.wikipedia.org/wiki/Metrology>).

also reduce error caused by any misalignment of the standard lamp in the goniophotometer. The standard lamp should also have a luminous intensity consistent with the signal level and system gain setting at which the goniophotometer will be used.

A luminous intensity standard lamp is mounted in the goniophotometer to reproduce the specific orientation and direction in which the lamp was calibrated. The photometric center of the standard lamp is positioned at the center of rotation of the goniophotometer.

7.2.1.2 Luminous Intensity Calibration Process. At the beginning of a calibration cycle, each standard lamp is operated in the goniophotometer. After the lamp has stabilized, readings are obtained using the goniophotometer photodetector along the normal optical path, inclusive of any system mirrors or other optical elements. The lamp electrical data should be recorded to verify that the standard lamp is operating under the specified conditions. The calibration factor, R_I , is obtained by:

$$R_I = i_s / I_s,$$

where:

R_I = the goniophotometer responsivity, A/cd or cts/cd

i_s = the photometer reading, A or cts

I_s = the luminous intensity of the standard lamp, cd

Multiple readings, with different standard lamps, are strongly recommended to establish one calibration factor. In such cases, the arithmetic mean of calibration factors for all standard lamps is employed as the calibration factor of the goniophotometer.

This calibration factor is specific to the system gain setting at which the calibration measurements were made. If the linearity of the system is sufficient, the factor is scaled in proportion to the gain ratio for other system gain settings. Normally, a silicon-photodiode-based detector with a well-designed electrical signal processing system has sufficient linearity for most measurements; however, the measured change in signal between system gain settings can be significant. Most transimpedance amplifiers, which convert photocurrent to voltage, have a linearity specification on the order of 1% between

system gain settings. The linearity between different gain settings should be investigated, and calibration factors should be established for each system gain setting.

An important geometrical concern arises when the light emitted from the luminous intensity lamp in the opposite direction of the photometer is reflected toward the photometer. In this case a bias in the calibration is created, which propagates through luminous intensity distribution measurements and total luminous flux measurements. This form of stray light can be minimized or nearly eliminated by using a black material with a specular surface to absorb and reflect the light away from the photometer, where it can be absorbed without affecting the measurement.

7.2.2 Illuminance Calibration

7.2.2.1 Illuminance Standard. An illuminance standard is a photometer that has been calibrated by a laboratory capable of establishing an absolute value of illuminance (lux, lx) at a plane, such as an NMI or a commercial calibration laboratory. Testing laboratories may also use a standard photometer internally calibrated for illuminance responsivity (signal per lux) where the scale is transferred from SI-traceable standards for their own use.

A standard photometer to be used in a goniometer shall have certain characteristics:

- The field of view shall encompass the entire luminaire under test.
- The angular responsivity of the standard photometer should closely approximate the cosine law over the angular region defined by the field of view.
- For goniometer designs in which the photometer is illuminated at varying angles, the standard photometer responsivity should be nearly uniform with respect to entrance angle.
- The standard photometer shall have a well-defined reference plane that follows the inverse-square law, and a spectral responsivity that closely matches the $V(\lambda)$ function. IES Lighting Measurement (LM) documents related to goniometric measurements specify the requirements, such as, f_1' which is a measure of the closeness to the $V(\lambda)$ function.

7.2.2.2 Illuminance Calibration Process. The goniophotometer calibration process with a standard photometer calibrated for illuminance responsivity is relatively straightforward. The standard photometer shall be mounted such that its reference plane is aligned with the reference plane of the goniometer detection system. The standard photometer aperture is centered along the optical axis, and the reference plane is perpendicular to the optical axis. This alignment is usually accomplished with a laser.

To use the standard photometer to make luminous intensity distribution measurements, the distance from the photometric center of the goniophotometer to the reference plane of the standard photometer shall be determined. A laser range finder usually provides the result with the lowest uncertainty.

If the standard photometer is mounted in a rotating-mirror goniometer, the reflectance of the mirror shall be accounted for in the calibration. A luminous intensity lamp may be used to determine the mirror reflectivity factor by determining the ratio of the luminous intensity lamp signal reflected off the mirror compared to the signal of the luminous intensity lamp measured at the same photometric distance without the use of a mirror. The measurement of the luminous intensity lamp without the use of the mirror will require mounting the standard photometer at the photometric distance of the goniophotometer and aligning the luminous intensity lamp to the new photometer position.

7.2.3 Total Luminous Flux Calibration

7.2.3.1 Total Luminous Flux Standard. A total luminous flux standard is a light source that has been calibrated by a laboratory capable of establishing an absolute value of total luminous flux (lumens, lm), such as an NMI or commercial calibration laboratory. Testing laboratories may also use internally calibrated working-standard lamps, where the total luminous flux is transferred from SI-traceable standards for their own use.

A high-quality total luminous flux standard lamp should have the following characteristics:

- As near an isotropic distribution as possible (to minimize sampling errors from the measurements,

which typically only include a percentage of the total solid angle)

- A correlated color temperature near 2856 K (CIE Standard Illuminant A), the CCT at which the photometer response is characterized
- Negligible variation in CCT with angular position
- Negligible loss of lumens due to shadowing of source from goniophotometer mechanical or structural elements

Calibrated incandescent (including halogen) lamps meet these requirements well, but the shadowing of the goniophotometer requires designed features in order to be minimized. In some cases, this could mean extending the length of the lamp holder stalk to reduce the angle of blocked light near 180 degrees vertical. Furthermore, calibrated flood-type halogen reflector lamps meet the criteria, but the isotropic distribution is not realized as well as for general service lamps, so tighter angular measurements may be required.

7.2.3.2 Total Luminous Flux Calibration Process. At the beginning of a calibration cycle, each standard lamp is operated in the goniophotometer. After the lamp has stabilized, measurements are made using a sweep of the goniophotometer photodetector in a set of circumpolar and planar angles. The appropriate sampling interval depends on how isotropic the standard lamp is (see **Section 8**). The lamp electrical data should be recorded to verify that the standard lamp is operating under the specified conditions. The calibration factor R_l is obtained by:

$$R_l = \frac{1}{\Phi_s} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} [i_s(\theta, \phi) - i_{s_d}(\theta, \phi)] \sin \theta \, d\theta \, d\phi,$$

where:

R_l = goniophotometer responsivity for each standard lamp, A/cd or cts/cd

Φ_s = total luminous flux of the standard lamp (lm)

$i_s(\theta, \phi)$ = the photometer signal for the measurement, A or cts

$i_{s_d}(\theta, \phi)$ = the dark reading (with shield in place), which includes a measurement of the stray light in the room, A or cts (see **Section 9**)

Multiple readings with different standard lamps are strongly recommended to establish one calibration factor. In such cases, the arithmetic mean of calibration factors for all standard lamps is employed as the calibration factor of the goniophotometer.

8.0 Integrated Measurements

8.1 Method of Calculation

An integrated measurement over a specified solid angle is simply the integration of smaller solid angles weighted by the measurement quantity. For example, the total luminous flux is calculated by using:

$$\Phi_v = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I(\theta, \phi) \sin \theta \, d\theta \, d\phi,$$

where:

- Φ_v = the total luminous flux of the lamp (lm)
- $I(\theta, \phi)$ = the luminous intensity (cd) of the lamp in a direction defined by a circumpolar and apolar angle
- θ = the circumpolar angle
- ϕ = the apolar angle

The luminous intensity for each measured direction is determined by:

$$I(\theta, \phi) = \left[i_s(\theta, \phi) - i_{sd}(\theta, \phi) \right] / R_I,$$

where:

- R_I = the goniometer calibration factor, expressed in the units of the signal (amperes, volts, or counts) per candela
- $i_s(\theta, \phi)$ = the photometer signal in a direction
- $i_{sd}(\theta, \phi)$ = the photometer dark (i.e., stray light) signal in a direction

The stray light signal is measured with the standard lamp operating but the direct light blocked. The direct light is typically blocked by placing a black cloth over the mirror. An ideal block would be a black cloth that only blocks the image of the lamp. This signal captures the electrical dark noise, along with stray light present in the room (see **Section 9**). Both photometer signals are normally measured as a current, voltage, or counts from an analog-to-digital converter.

To evaluate the integral, a summation is used to approximate the final integrated quantity as shown by:

$$\Phi_v = 2\pi \sum_{\theta=0}^{\pi} \left[\left(\frac{1}{n} \sum_{\phi=0}^{2\pi} I(\theta, \phi) \right) \sin \theta \Delta\theta \right]$$

For the IES coordinate systems, the summations⁵ are:

IES-A & IES-B:

$$\Phi_v = \frac{2\pi}{M} \sum_{m=1}^M \left\{ \sum_{n=-N+1}^N I\left(\frac{2\pi m}{M}, \frac{\pi n}{2N}\right) \left[\sin\left(\frac{n\pi}{2N}\right) - \sin\left(\frac{(n-1)\pi}{2N}\right) \right] \right\}$$

IES-C:

$$\Phi_v = \frac{2\pi}{M} \sum_{m=1}^M \left\{ \sum_{n=1}^N I\left(\frac{2\pi m}{M}, \frac{\pi n}{N}\right) \left[\cos\left(\frac{(n-1)\pi}{N}\right) - \cos\left(\frac{n\pi}{N}\right) \right] \right\},$$

where M is the number of half-planes, and N is the number of planar angles measured.

8.2 Angular Frequency of Data Points

The necessary angular resolution of goniophotometric measurements depends on the structure of the luminous intensity distribution, the intended end use of the data, the required measurement uncertainty, and any a priori knowledge of the nature of the luminous intensity distribution at relatively high angular frequencies. A luminous intensity distribution should be sampled at an angular frequency such that the measured luminous intensity varies smoothly and predictably between data points. An unpredictable spike in the luminous intensity distribution between data points may cause problems for the end user in final application of the test luminaire.

For integrated values, it is critical that the luminous intensity distribution between two data points be smooth and predictable. The total luminous flux equation uses the measured data point to represent the average luminous intensity over the solid angle. For a typical mirror goniometer, the collected solid angle is on the order of 3.5×10^{-7} sr (for a photometer with an aperture of 1 cm on a 15-m system), which is used to represent a solid angle of 3.4×10^{-5} sr (measured at the equator of the system, for a step size of 5° for the circumpolar and planar angles). The measured value directly represents only 1% of the solid angle being sampled.

Many application-specific light measurement test methods state specific angular frequencies of measurement for

both circumpolar and planar angles.⁶⁻¹⁰ Good laboratory practice prescribes that the testing laboratory assess the adequacy of these suggestions, either visually or through analysis. Analysis of the distribution should be performed to determine the number of lateral and vertical angles required to adequately define the distribution.

9.0 Stray Light Elimination and Correction

9.1 Stray Light and Its Sources

Stray light is light measured by the goniophotometer detector that does not follow the intended path from the luminaire to the detector. The light from the luminaire can scatter off the walls, floor, and/or ceiling and into the goniophotometer detector, either directly or after being reflected from the luminaire or the mirror.

9.2 Stray Light Elimination Techniques

Elimination of stray light involves the physical modification of the goniophotometer or its environment to reduce the amount of stray light reaching the goniophotometer detector. Many methods exist for eliminating stray light.

9.2.1 Limiting Field of View. The goniophotometer detector should be designed to limit its field of view as much as possible, without clipping the image of the luminaire. The use of a diffuser on a goniophotometer detector reduces angular effects when measuring a large luminaire, but makes the goniophotometer more susceptible to stray light because it will accept light from any direction. Limiting the field of view is typically done using apertures and baffles arranged in such a way as to not allow light scattered in the room to reach the goniophotometer detector. With a rotating-mirror goniophotometer, the aperture size can be minimized by placing the aperture approximately half a meter in front of the goniophotometer detector and rotating it synchronously with the mirror.

9.2.2 Reducing Reflected Light. Reducing reflected light from all surfaces in the room will significantly reduce the amount of stray light. The room surfaces and structures holding the mirrors, and the edges of the mirror itself, should be covered with a diffuse (matte) black finish, such as that provided by blackout materials

and high quality black felt curtains. Care shall be taken to avoid specular reflections between the source and the detector, even if the surfaces are black. Baffles along walls, ceilings, and even floors can be employed to reduce stray (particularly specular) reflections.

9.2.3 Use of a Light Trap. For best results, a light trap should be positioned opposite the detector, behind the luminaire, and rotated synchronously with the moving mirror or goniophotometer detector. A light trap is typically made using several surfaces composed of black material with a specular finish, which are placed at acute angles (approximately 45°) to each other and tilted slightly off the optical axis. The light enters the trap and is mostly absorbed as it is bounced from the first surface to a second surface. The process repeats for many reflections until virtually all the light is absorbed, as shown in **Figure 9-1**. While extremely effective, such light traps are not commonly used in goniophotometry.

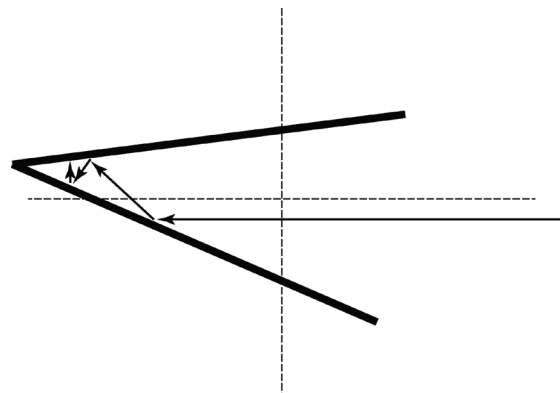


Figure 9-1. Multiple bounces within a slightly tilted light trap.

9.3 Stray Light Correction Methodologies

Correction of stray light is the mathematical subtraction of measured signal based on knowledge of the goniophotometer system, or on additional, stray light measurements. Stray light measurements may be made by placing a baffle or mask between the luminaire and the goniophotometer detector to block all direct rays of light, and then repeating the full measurement.

9.3.1 Using a Mask. A *mask* made of black diffusing material, such as black felt, may be placed on the mirror to block the light passing directly from the luminaire to the goniophotometer detector. The mask should be large enough to cover the image of the luminaire, yet small

enough to minimize the amount of blocked stray light from surfaces in the background. Stray light is then measured and subtracted from the photometric test data acquired using one of these procedures:

- **Single or multiple polar angle stray light measurement procedure:** If the luminaire is quite symmetrical in luminous intensity distribution, a stray light signal may be measured at one polar angle that is above the horizontal. For luminaires that have an asymmetric luminous intensity distribution, stray light measurements should be made at multiple polar angles (a minimum of three) and then averaged. The stray light signal is then subtracted from all the measurements.
- **Single or multiple vertical-plane stray light measurement procedure:** The most rigorous method is to acquire a full set of stray light data at the same set of angular coordinates that are used to test the luminaire. After the full set of stray light data is collected, each stray light reading is subtracted from the corresponding luminous intensity reading collected for the luminaire under test. In many cases, this method is not practical, since it doubles data acquisition time. A more practical method is to collect stray light data in a single polar plane, and then to treat the result as representative of the entire stray light distribution, subtracting these values from the luminous intensity readings acquired in each polar plane for the device under test. This method is appropriate when the characteristic stray light distribution of the goniophotometer system has been shown to be approximately uniform with respect to circumpolar angle.

9.3.2 Handling of Negative Luminous Intensity Values. The process of subtracting stray light readings may result in negative values for some of the corrected luminous intensity measurements. Such negative values are due to the influence of random noise when the measurand approaches zero. Random noise is composed of both negative and positive values, which tend to cancel—i.e., average to zero—during angular integration. If the negative luminous intensity values were arbitrarily removed, the residual noise would no longer be random, and would not tend to cancel when integrated quantities were calculated. Instead, it would introduce a positive bias in the measurement result.

The appropriate handling of negative luminous intensity values depends on the intended application of the resulting data. When the data are to be used to calculate integrated flux quantities, negative values shall be retained. When the data are to be used exclusively to represent the angular distribution of luminous intensity, negative values may be set to zero. When measurement results are to be used for both purposes, it may be necessary to record these results in two different ways. When the data are recorded in the standard IES file format, specific labeling requirements apply:

- For IES files that are to be used for calculating integrated flux quantities, any negative values shall be retained, and a line in the header code shall be added stating specifically:

```
[OTHER] For integrated flux
calculations - may contain negative
candela values
```
- For all other IES files, any negative values shall be set to zero.

9.4 Stray Light and Uplight Measurement

9.4.1 Uplight Detection Threshold. Even with the most rigorous application of the techniques described above, stray light and photometric noise will still contribute to the measured values in a goniophotometric exploration. This can cause positive non-zero values to be measured at angles where the luminaire is not emitting light. These positive non-zero values are problematic when the test results are intended to verify compliance with specifications that require zero light at certain angles (notably, the BUG ratings of the Luminaire Classification System, described in IES TM-15-11¹¹).

For example, the luminaire enclosures depicted in **Figure 9-2** are designed with a horizontal flat light-emitting opening, with no external light-directing devices. The design of this type of luminaire makes it impossible for light to be directed at or above 90 degrees vertical when the luminaire is mounted with the luminous opening horizontal. Because of the presence of stray light in the goniophotometer system, positive non-zero readings may still be recorded at these angles. If these spurious readings were reported as positive non-zero luminous intensity values, the luminaire could not be classified as “U0” according to TM-15-11.

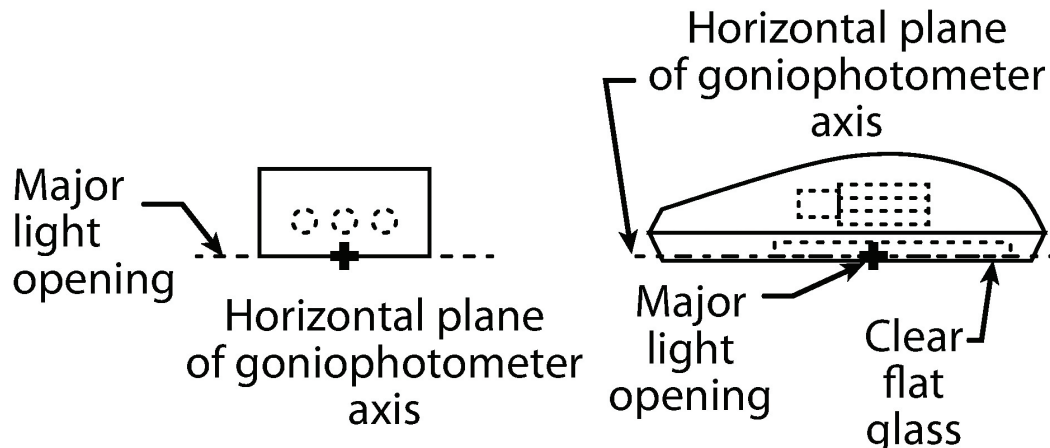


Figure 9-2. Luminaire with horizontal flat light-emitting opening.

To avoid such misclassification of a luminaire, it is necessary to define the applicable test in terms of *detection*, rather than *measurement* of uplight. While it is impossible to verify that the uplight from any luminaire is exactly zero, it is possible to verify that a luminaire emits no uplight detectable by a given measurement system. This necessarily involves determining the detection threshold of the measurement system, and the comparison of the apparent uplight measured for the luminaire with that threshold.

For a luminaire designed to achieve zero uplight, which is submitted for testing to support U0 classification, positive non-zero values that exist in the test data at and above 90V may be ignored, or assigned a zero value, only if analysis of the data collected at and above 90V is consistent with the design-based assumption that the device under test produces zero uplight. This is the case when the apparent uplight is below the detection threshold of the measurement system, and is attributable to the effects of stray light and noise in the detector signal.

Determination of the uplight detection threshold of a given measurement system involves consideration of various system characteristics, including stray light and photometric signal-to-noise performance. To ensure reproducibility of test results, a *conventional goniophotometer system* is defined as one which has an uplight detection threshold equal to 0.25% of the total lumen output of the luminaire. Any measurement system used to perform U0 verification testing shall be

shown to have an uplight detection threshold less than or equal to that of the conventional goniophotometer system. More-specific criteria are enumerated in **Section 9.4.2**. For this Standard, *detectable uplight* is defined as uplight that is detectable by a conventional goniophotometer system.

9.4.2 Criteria for Zeroing Uplight Readings. Once stray light elimination and correction methodologies have been implemented, it may be reasonable to *zero* (i.e., to assign a zero value to) data in the upper hemisphere under certain circumstances, as discussed in **Section 9.3.2**. The laboratory shall be allowed* to zero upper hemisphere data that falls below the threshold of 0.25% of the total luminaire luminous flux, when these two criteria are met:

1. The laboratory has demonstrated the ability to *eliminate* stray light to a level less than 1% of the total luminaire luminous flux under the test condition described in **Section 9.4.3**.
2. The laboratory has demonstrated the ability to *correct* for stray light to a level less than 0.1 % under the test condition described in **Section 9.4.3**, and such correction is implemented for the measurement of the luminaire under test.

9.4.3 Stray Uplight Test Setup. The test condition used to meet the requirements of **Section 9.4.2** involves the measurement of a 500-watt clear, T4, G9.5 medium

* The laboratory should determine that the distribution of any signal being zeroed is consistent with the typical stray light distribution for the goniophotometer measurement system.

bi-pin based tungsten halogen lamp, operated at the rated voltage in a base-up orientation, and mounted within a cylindrical structure 15 cm (6 in.) in diameter. The lamp shall be centered within the cylinder and positioned such that the tip of the lamp is recessed between 1 mm and 5 mm from the plane formed by the open bottom of the cylinder. The top of the cylinder shall be completely enclosed and not allow any light to escape. Vent tubes may be added to allow heat to leave the top of the cylinder, but the vent tubes shall exhaust at the level of the open end of the cylinder. The inside and the outside of the cylinder shall be painted with high-temperature matte black paint to minimize light reflection. The first axis of the lamp shall be aligned with the polar axis of the goniometer, and the plane formed by the open end of the cylinder shall be aligned with the photometric center of the goniometer.

9.4.4 Stray Uplight Test Procedure. The test shall be conducted following LM-46-04.¹¹ Two tests are performed, with and without stray light correction, in lateral increments of 22.5° (16 half-planes) and vertical increments of 2.5° from 0° to 180°:

Uplight measurement without stray light correction: The first test involves no stray light correction measurements.[†] For the purposes of this test, *stray uplight* is any light measured above the horizontal plane defined by the open end of the cylinder. To qualify the system, the uncorrected stray uplight shall not exceed 1% of the total luminous flux measured.

Uplight measurement with stray light correction: The second test involves stray light correction. To qualify the system, the stray uplight for the corrected measurement shall not exceed 0.1% of the total luminous flux.

Stray light characterization: Stray light has a pattern that is dependent on the goniophotometer, its placement in the laboratory, and the reflectances of the surfaces of the goniophotometer and the surrounding environment. Stray light may be a

constant signal (a uniform glow) due to the scattering in the room, or stray light may have specific pattern, such as a peak opposite the angle of peak intensity for a given luminaire, or a peak opposite a feature in the room (a control box or mechanical parts of the goniophotometer). The laboratory shall analyze the results of the tests described in **Sections 9.4.4.1** and **9.4.4.2** and determine the normal patterns for their goniophotometer system to meet requirement #3 in **Section 9.4.2**.

9.5 Reporting Zeroed Stray Uplight

For a luminaire submitted for BUG testing, if the three criteria enumerated in **Section 9.4.2** are met, then luminous intensity values for angles at and above 90°V may be omitted from the test report, or assigned a zero value (refer to the application-specific light measurement test method; e.g., the applicable current version of IES LM-10, LM-31, or LM-79). In such a case, the original data shall be retained for reference, and the details of the supporting detection threshold analysis shall be documented. In this case, the test report shall include the following statement:

*The device under test emits no detectable uplight,
as defined by ANSI/IES LM-75-19.*

If uplight values are assigned a zero value, the test report shall also include the statement:

*For the purpose of this report, certain non-zero uplight
readings have been assigned a zero value, in accordance
with the requirements of ANSI/IES LM-75-19*

For IES files that represent luminaires where data have been zeroed based on the stray light analysis presented in this document, a line in the header code shall be added stating specifically:

[OTHER] The reported results were measured in a facility that meets the requirements of ANSI/IES LM-75-19. The device under test emits no detectable uplight, as defined by ANSI/IES LM-75-19.

[†] For systems where the amplifier output is intentionally set to provide a positive reading even when no light is present, a stray-light plane should be measured with the light off.

Annex A – Coordinate System Conversion

A.1 Purpose

This Annex presents a procedure for calculating coordinate transformations between IES Coordinate Systems A, B, and C. Analogous transformations can be made between the three standard CIE Coordinate Systems, or between IES and CIE Coordinate Systems, by simply substituting CIE coordinates (A, α) for IES coordinates (Y, X) , (B, β) for (V, H) , and (C, γ) for (V, L) .

A.2 Direct Transformation between Specific Standard Coordinate Systems

Because the transformations defined below involve arctangent functions, some problems may occur in implementation unless appropriate care is taken in programing. To mitigate the problems in programming, the atan2 function is used, and is defined as follows. (Note that the angle is expressed in radians for the atan2 definition, while it is expressed in degrees for the conversion equations.)

$$\text{atan2}(y, x) = \begin{cases} \tan^{-1}\left(\frac{y}{x}\right) & \text{if } x > 0 \\ \tan^{-1}\left(\frac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \tan^{-1}\left(\frac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ +\frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \\ \text{undefined} & \text{if } x = 0 \text{ and } y = 0 \end{cases}$$

IES-A → IES-B:

$$V = \sin^{-1}(\cos Y \sin X)$$

$$H = \begin{cases} 0 & \text{if } \sin Y = 0 \text{ and } \cos Y \cos X = 0 \\ \text{atan2}(\sin Y, \cos Y \cos X) & \text{elsewhere} \end{cases}$$

IES-A → IES-C:

$$V = \cos^{-1}(\cos Y \cos X)$$

$$L = \begin{cases} 0 & \text{if } \sin X \cos Y = 0 \text{ and } \sin Y = 0 \\ \text{atan2}(\sin Y, \sin X \cos Y) & \text{else if } \sin Y \geq 0 \\ 360 + \text{atan2}(\sin Y, \sin X \cos Y) & \text{elsewhere} \end{cases}$$

IES-B → IES-A:

$$Y = \sin^{-1}(\cos V \sin H)$$

$$X = \begin{cases} 0 & \text{if } \sin V = 0 \text{ and } \cos H \cos V = 0 \\ \text{atan2}(\sin V, \cos H \cos V) & \text{elsewhere} \end{cases}$$

IES-B → IES-C

(subscripts added to minimize confusion):

$$V_C = \cos^{-1}(\cos V_B \cos H)$$

$$L = \begin{cases} 0 & \text{if } \sin V_B = 0 \text{ and } \sin H \cos V_B = 0 \\ \text{atan2}(\sin H \cos V_B, \sin V_B) & \text{else if } \sin B \cos V_B \geq 0 \\ 360 + \text{atan2}(\sin H \cos V_B, \sin V_B) & \text{elsewhere} \end{cases}$$

IES-C → IES-A:

$$Y = \sin^{-1}(\sin V \sin L)$$

$$X = \begin{cases} 0 & \text{if } \cos L \sin V = 0 \text{ and } \cos V = 0 \\ \text{atan2}(\cos L \sin V, \cos V) & \text{elsewhere} \end{cases}$$

IES-C → IES-B (subscripts added to minimize confusion):

$$V_B = \sin^{-1}(\sin V_C \cos L)$$

$$H = \tan^{-1}(\tan V_C \sin L)$$

$$X = \begin{cases} 0 & \text{if } \sin L \sin V_C = 0 \text{ and } \cos V_C = 0 \\ \text{atan2}(\sin L \sin V_C, \cos V_C) & \text{elsewhere} \end{cases}$$

Annex B – Zonal Constant Calculations

The zonal constant is a factor by which the mean intensity emitted by a source of light in each angular zone is multiplied to obtain the lumens in the zone. These constants may also be used to calculate the lumens in any sector, and thus the total lumens from a luminaire. Care should be taken to choose areas small enough to approach integration.

By multiplying the luminous intensity values by the appropriate constants, lumens for the corresponding segment of each zone are obtained. Addition of all lumen values corresponding to a given angular range will supply the lumens in the total zone represented.

For computer calculation, the zonal constants can be expressed by:

$$K = \frac{\pi(\cos\theta_1 - \cos\theta_2)(\phi_1 - \phi_2)}{180},$$

where θ_1 and θ_2 are angles from the polar axis (in degrees) bounding the angular zone, and ϕ_1 and ϕ_2 are circumpolar angles (in degrees) bounding the angular zone.

A simplified calculation may be made for applications where only the total lumens for a given apolar zone are of interest, such as when calculating bare lamp output. For each apolar angle, a weighted average of the test lamp's relative luminous intensity for all circumpolar angles shall be determined. The zonal constants when calculated for a total apolar zone can be calculated with the simplified equation:

$$K = 2\pi(\cos\theta_1 - \cos\theta_2)$$

ANNEX C – INFORMATIVE REFERENCES

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